

SAGAR MANDIYA

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EDUCATION

Boston University

Boston, MA

Master of Science in Computer Science (STEM)

Jan 2024

Relevant Coursework: Artificial Intelligence, Machine Learning, Software Engineering, Analysis of Algorithms, Data Science with Python, Web Mining and Analytics

SKILLS & CERTIFICATION

Technical Languages: Python, C, C++, R, MATLAB, SQL, JavaScript, HTML, CSS

Frameworks: TensorFlow, PyTorch, CUDA, TensorRT, Nvidia Riva, Nvidia Triton, Nvidia DeepStream, FastAPI, Django, React

Developer Tools: Git, Docker, WordPress, Bootstrap, Oracle Cloud Platform, AWS, GCP, LaTeX

Operating Systems: Linux, MacOS, Windows

Certifications: Deep Learning Specialization, Convolutional Neural Networks in TensorFlow, Machine Learning

EXPERIENCE

Software Engineer Intern

Boston, MA

Ipser Lab

May 2024 – Present

- Developed React based components for a business website **to reduce load times and improve user experience**.
- Utilized Java to **resolve login issues** at the back-end to ensure **consistent performance across distinct platforms**.
- Actively contributed in brainstorming ideas with the managing team to crucially improve the overall product performance.

Graduate Research Assistant

Boston, MA

Boston University - Department of Computer Science

Mar 2023 – Mar 2024

- Analyzed ER (Emergency Room) data using data science and machine learning techniques, deriving insights to **minimize press machine related work injuries** to an estimate of **more than 20%**.
- Generated compelling **data visualizations using Python and Tableau**, **reducing injury misclassification by 15%** and enabling more accurate injury severity and recovery time predictions.
- Utilized ML models and dimensionality reduction to analyze children with autism spectrum disorder (ASD), for one of the **largest longitudinal studies** (>15000 samples) of **first language acquisition in autistic children**.

Computer Vision Engineer

TensorGo Technologies

Sep 2021 – Aug 2022

- Designed and built virtual perimeter overlay, utilizing **image segmentation and object detection** to replace soccer field boundary advertisements with more relevant ads resulting in **targeted audience marketing**.
- Executed innovative **image processing** techniques on **UNet++** to optimize virtual perimeter overlay, attaining a **30% reduction in processing time while maintaining similar accuracy**.
- Led a team as project manager to build and deploy a comprehensive pipeline for **BERT based language model from scratch**, **reducing hiring time for recruiters by up to 25%**.
- Built a complete pipeline to facilitate end-to-end ASR (**Automatic Speech Recognition**), **Voice Biometrics**, and **Diarization** utilizing Nvidia NGC stack (**CUDA, RIVA, NEMO, Triton**), resulting in **20% faster model**.
- Formulated and managed projects collaboratively, addressing issues, and strategically engaging with team members and stakeholders.

PROJECTS

AutoCode for Python | Python, PyTorch, NLTK, ONNX, GCP, Google BigQuery, SQL

- Trained Transformer based GPT-2 based LLM model from scratch to auto-complete python code similar to GitHub CoPilot.
- Employed Google BigQuery to gather a large 160GB python code dataset from Github to be used to train the LLM.

Deepfake Detection | Python, TensorFlow, TensorRT, XceptionNet, MisoNet, ResNet, VGG-19

- Designed and implemented neural network models for **detecting artificially generated fake images/videos** with 73.5% average precision on a **large dataset of size 470GB+**.
- Conducted a comparative study of SOTA neural network models to determine performance for deepfake detection. [[paper](#)]

PUBLICATIONS

Analyzing Patterns of Injury in Occupational Hand Trauma Focusing on Press Machines: Registry Based Study and Machine Learning Analysis [Accepted Springer]

May 2023 – Nov 2023

Critical period for first language acquisition may be shorter in autistic children than in typical children [[paper](#)]

July 2023 – Jan 2024